

Expression of Interest

Blueprint Biosecurity — Request for Proposals: Pathogen-Agnostic Biothreat Detection

Technical Area: 3 — Modeling pathogen-agnostic biosurveillance system features such as cost, sensitivity, and optimal deployment

Project Title: Days per Dollar (DPD) — Sequential Detection Statistics and Calibrated End-to-End Lead-Time Modeling for Metagenomic Biosurveillance Deployment

Principal Investigator: Sihua Peng, PhD — Research Scientist, College of Public Health, University of Georgia; affiliated with the Georgia Advanced Computing Resource Center (GACRC)

Institution: University of Georgia, Athens, GA, USA

Requested Period of Performance: 12 months

Approximate Budget Request: \$267,300 (direct + indirect; see Section 6)

Date: [INSERT] | **Contact:** [INSERT EMAIL / PHONE]

1. Problem Statement

Every operational biosurveillance program faces the same budget meeting: someone asks what another million dollars buys, and no one can answer in days of warning. The question is not rhetorical — it determines site counts, sampling cadence, and sequencing depth for CDC Biothreat Radar, NWSS, TGS, and their international counterparts — and it is currently answered by intuition.

Two gaps make it unanswerable. First, **the detection endpoint itself is undefined**. Cost-sensitivity models must assume a bioinformatic detection threshold, because no one has characterized one operationally: three reads at 10^{-7} relative abundance in one sample cannot be distinguished from noise, and essentially all deployed MGS analysis is single-sample, discarding the time dimension where the signal actually lives. An agent growing exponentially produces a *trajectory*, and a trajectory is detectable long before any single timepoint clears a threshold. Second, **the modeling chain is broken at the joints**. Shedding models, P2RA relative-abundance factors, cost models, and siting analyses exist as separate literatures with incompatible parameterizations, so no one can propagate uncertainty from a shedding parameter through to a deployment decision.

DPD closes both. We will build a sequential detection framework whose *measured* operating characteristic — detection delay at a fixed false-alert rate — becomes the calibration input to a differentiable end-to-end simulator that maps budget directly to lead time, then validate the whole chain against events that actually happened.

2. Technical Approach

Aim 1 — Anytime-valid sequential detection on MGS time series (Months 1–7)

We will treat detection as a **sequential change-point problem over thousands of parallel hypotheses**, not a per-sample classification.

Observation model. Read counts for each sequence cluster (k-mer cluster or contig cluster) are modeled as negative-binomial / Dirichlet-multinomial conditional on library depth, with explicit terms for run and batch effects, fecal-strength normalization (PMMoV, crAssphage), flow and dilution covariates, and extraction-control efficiency. Compositionality is handled directly rather than assumed away — a taxon's relative abundance can fall while its absolute load rises, and a normalization choice that ignores this manufactures both false alarms and false reassurance. We will systematically compare normalization strategies, which is itself an open question the RFP names.

Sequential test. Bayesian online change-point detection and CUSUM on log-abundance trajectories, plus **e-value / test-martingale sequential tests that are anytime-valid**. This last point is methodologically load-bearing: an operational program looks at its data every week, and classical fixed-sample FDR control is simply invalid under repeated looks. E-processes with e-BH multiplicity control give valid inference at every peek, at a quantified cost in power that we will report rather than hide. Site-specific empirical nulls are built from historical baselines so that thresholds are calibrated to each catchment rather than imported.

Primary metric. Not AUC. We report the **detection-delay distribution at a fixed false-alert rate** — the surveillance analogue of average run length — because that is the quantity an operator and a funder can both act on. We additionally deliver growth-rate and doubling-time estimates with honest uncertainty, directly answering the RFP's question of a taxon's trajectory.

Data. NWSS (PRJNA747181) and CASPER (PRJNA1247874) time series as the primary substrate, Tisza et al. 2023 (PRJNA966185) and Wolfe et al. 2026 (PRJNA1438722) for capture-based contrast, and Zephyr/Broad pooled nasal swabs (PRJNA1052714) to test whether the framework transfers across sample types with different shedding and pooling structure.

Aim 2 — leadtime: a differentiable end-to-end budget-to-lead-time simulator (Months 3–11)

We will implement the full causal chain as a single differentiable JAX pipeline: pathogen introduction → epidemic growth (branching process with overdispersion k , plus SEIR comparison) → sample-type-specific shedding kinetics → catchment capture and collection cadence → wet-lab efficiency including extraction, library complexity, and duplication → sequencing depth and read allocation → **bioinformatic sensitivity curve calibrated by Aim 1's measured operating characteristic** → alerting rule → **lead-time distribution relative to status-quo syndromic detection**.

The coupling to Aim 1 is the point of the project. Existing cost models must *assume* a detection threshold; ours *measures* one on real wastewater and real pooled swabs, so the resulting dollar-to-days curve rests on an empirical foundation rather than a plausible-sounding constant.

We build on and extend Grimm et al. 2025 (P2RA), generalizing a point relative-abundance factor into a time-resolved, uncertainty-propagating quantity, and cross-check against EpiSewer- and wwinference-style formulations adapted to the compositional MGS setting. Because the pipeline is differentiable, budget allocation becomes gradient-based constrained optimization over site count, cadence, depth per sample, and sample-type mix — rather than the coarse grid search that current practice permits.

Parameter uncertainty is treated as a deliverable, not an embarrassment: global sensitivity analysis via Sobol indices identifies which parameters actually move the lead-time answer, telling Blueprint where measurement investment would most reduce decision uncertainty. Outputs are intervals; we will decline to publish point estimates that the data do not support.

Aim 3 — Empirical deployment optimization and retrospective validation (Months 5–12)

A national optimizer with no real constraints is a toy. We ground the model in one real region.

Integrating NWSS site metadata, census demographics, LODES commuting flows, Hartsfield–Jackson passenger volumes (the world's busiest hub and a CDC TGS site), and hospital referral networks, we formulate joint maximum-coverage siting and pooling-strategy optimization for Georgia and the Southeast, producing a framework any state can re-parameterize.

Retrospective validation asks the only question that settles the matter: **would this have caught it, and how much earlier?** We test the optimized site set plus Aim 1's alerting rule against mpox 2022 and H5N1-in-wastewater 2024 in archived public data. To prevent hindsight bias, the comparison definition — reference detection dates, alert criteria, and success thresholds — will be written and timestamped before the retrospective analysis is run.

3. Deliverables and Open Availability

Three open-source packages released under MIT/Apache-2.0 on GitHub from month 3, developed in the open rather than dumped at project end: **mgs-sentry** (Aim 1 sequential detection, R and Python, nf-core-compatible), **leadtime** (Aim 2 JAX simulator with an interactive dashboard), and **sitewise** (Aim 3 siting optimizer). Calibrated parameter sets, benchmark time series, and all retrospective analysis code go to Zenodo with a persistent DOI. We anticipate two to three peer-reviewed publications, each preprinted at submission, plus a plain-language policy brief written for program managers rather than methodologists — the audience that actually sets sampling cadence. All outputs are usable independently of one another and independently of any tool we build.

4. Speed of Execution and Risk Posture

The project requires no wet-lab work, no new sample collection, and no new hardware. Every primary dataset is already public and already downloaded by our group. Analysis begins in week one, and the 12-month timeline carries genuine slack rather than optimism. Aim 1 is method development on data in hand; Aim 2 is simulation; Aim 3's only external dependency — site-level metadata beyond what is public — is initiated in month 1 and, if it stalls, Aim 3 degrades gracefully to fully public NWSS site data without losing the retrospective validation.

5. Team and Capability

The PI has 40+ peer-reviewed publications (h-index 24) spanning computational biology and biostatistics, with formal training and publication record in statistical methodology and experimental design alongside current production metagenomics work across Illumina, PacBio HiFi, and Nanopore platforms. Current pipelines run at scale on the GACRC SLURM cluster via Nextflow and Apptainer, using sourmash, geNomad, VirSorter2, and Kraken2/Bracken. The combination that matters here is rarer than either half: sequential-testing and multiplicity-

control statistics on one side, and hands-on operational MGS pipeline engineering on the other, which is what allows Aim 2's cost model to be calibrated by Aim 1's measurements rather than by literature guesses.

GACRC provides all compute, storage, and container infrastructure; no new hardware is requested. We welcome Blueprint's introduction to operational partners (NWSS-participating utilities, Georgia DPH, CASPER, Zephyr) for parameterization and validation, and are open to teaming with groups submitting complementary TA1 or TA2 proposals — the leadtime sensitivity curve is a natural integration point for any TA1 detection method.

Companion submission disclosure: our group is separately submitting a TA1 EOI (**NOVA-MGS**, reference-free novelty scoring and agentic triage). The two are scientifically complementary — NOVA-MGS asks whether a sequence is novel, DPD asks when a trajectory warrants an alert and what that capability costs — but each is fully self-contained, independently scoped, and independently budgeted. Neither depends on the other being funded.

6. Budget Summary (approximate, 12 months)

Category	Amount
PI effort (30% FTE)	\$47,000
Postdoctoral researcher / computational scientist (100% FTE)	\$78,000
Graduate research assistant (50% FTE, incl. tuition)	\$46,000
Fringe benefits	\$38,000
Compute, storage, and GPU cloud burst capacity	\$22,000
Data acquisition, publication, dissemination, travel	\$12,000
Total direct	\$243,000
Indirect costs (10% of direct, per Blueprint policy)	\$24,300
Total requested	\$267,300

Indirect costs are budgeted at Blueprint's published 10% cap; the University of Georgia's negotiated F&A rate exceeds this, and we are initiating the institutional waiver request through the Office of Research now rather than at Full Proposal stage. The project is modular: **Aims 1 and 2 constitute a coherent standalone effort at approximately \$180,000** should Blueprint prefer a tighter scope, and Aim 1 alone is deliverable at approximately \$110,000.

7. Risks and Mitigations

Anytime-valid sequential tests are conservative. E-processes trade power for validity under continuous monitoring. We will benchmark e-BH against Benjamini–Yekutieli and fixed-sample controls on identical data, report the power cost in explicit terms, and ship both options so programs can choose their own operating point.

Parameter uncertainty in the end-to-end model is large. This is the honest state of the field, not a flaw we can engineer away. Mitigation is structural: Sobol global sensitivity analysis makes uncertainty a first-class output, and the model reports which unmeasured parameters most limit confidence — itself a useful finding for Blueprint's future funding priorities.

Retrospective ground truth is imperfect. Historical detection dates for mpox and H5N1 are themselves fuzzy and partly retrospective. We mitigate by pre-registering comparison definitions before analysis, reporting results under multiple defensible date conventions, and stating limitations in the main text rather than a footnote.

Cross-sample-type transfer may fail. Wastewater and pooled nasal swabs differ in shedding, pooling depth, and background. If the Aim 1 framework does not transfer cleanly, that is a reportable scientific result and the wastewater deliverable stands unaffected.

Submitted in response to Blueprint Biosecurity's Pathogen-Agnostic Biothreat Detection RFP. Contact: [EMAIL], [PHONE].